CSE STUDY HUB

Emerging Technology

UNIT 6 · Security & Bio-Engineering Domains

What this unit is about

This unit builds understanding of **Security & Bio-Engineering Domains**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Cybersecurity foundations: confidentiality, integrity and availability

2. Threats in connected systems: malware, spoofing, data theft and denial of service

3. Security by design, authentication, encryption and updates

4. IoT/CPS risk management, segmentation and monitoring

5. Bioengineering and biotechnology: core concepts and applications

6. Bioinformatics: biological data, sequence analysis and computational tools

7. Ethical, privacy, safety and societal issues in bioengineering

8. Responsible governance and interdisciplinary risk assessment

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 6 NOTES · PART 1

Cybersecurity foundations: confidentiality, integrity and availability

For Cybersecurity foundations: confidentiality, integrity and availability, discuss the objective, stakeholders and possible harm. Cover consent/authority, confidentiality, integrity, accountability, transparency and applicable policy or law. Recommend safeguards and explain how they are checked.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Threats in connected systems: malware, spoofing, data theft and denial of service

Threats in connected systems: malware, spoofing, data theft and denial of service is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Security by design, authentication, encryption and updates

For Security by design, authentication, encryption and updates, discuss the objective, stakeholders and possible harm. Cover consent/authority, confidentiality, integrity, accountability, transparency and applicable policy or law. Recommend safeguards and explain how they are checked.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Security & Bio-Engineering Domains, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 6 NOTES · PART 2

IoT/CPS risk management, segmentation and monitoring

IoT systems connect physical devices containing sensors/actuators to networks and software. A typical flow is sensing → connectivity/gateway → processing → application/action; security and interoperability are key concerns.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Bioengineering and biotechnology: core concepts and applications

IoT systems connect physical devices containing sensors/actuators to networks and software. A typical flow is sensing → connectivity/gateway → processing → application/action; security and interoperability are key concerns.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Bioinformatics: biological data, sequence analysis and computational tools

Bioinformatics: biological data, sequence analysis and computational tools is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Security & Bio-Engineering Domains, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 6 NOTES · PART 3

Ethical, privacy, safety and societal issues in bioengineering

Bioengineering applies engineering principles to biological systems, devices and processes. Evaluate safety, privacy, consent, fairness and environmental effects alongside technical benefits.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Responsible governance and interdisciplinary risk assessment

For Responsible governance and interdisciplinary risk assessment, discuss the objective, stakeholders and possible harm. Cover consent/authority, confidentiality, integrity, accountability, transparency and applicable policy or law. Recommend safeguards and explain how they are checked.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Security & Bio-Engineering Domains, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Cybersecurity foundations: confidentiality, integrity and availability** — know the definition, core working and one example.

• **Threats in connected systems: malware, spoofing, data theft and denial of service** — know the definition, core working and one example.

• **Security by design, authentication, encryption and updates** — know the definition, core working and one example.

• **IoT/CPS risk management, segmentation and monitoring** — know the definition, core working and one example.

• **Bioengineering and biotechnology: core concepts and applications** — know the definition, core working and one example.

• **Bioinformatics: biological data, sequence analysis and computational tools** — know the definition, core working and one example.

• **Ethical, privacy, safety and societal issues in bioengineering** — know the definition, core working and one example.

• **Responsible governance and interdisciplinary risk assessment** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Security & Bio-Engineering Domains. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Cybersecurity foundations: confidentiality, integrity and availability and explain its importance.

Q2. Explain the working/steps involved in Threats in connected systems: malware, spoofing, data theft and denial of service.

Q3. Differentiate between two related concepts from Security & Bio-Engineering Domains.

Q4. Give one practical application of Responsible governance and interdisciplinary risk assessment and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Security & Bio-Engineering Domains.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.